

Java divides the into the following groups:

- Arithmetic operators
- Assignment operators
- Comparison operators
- Logical operators
- Bitwise operators

+ , - , * , / , %

Variables

int

x = 5.5

x = 7

x = Ram

Arithmetic Operators

Arithmetic operators are used to perform common mathematical operations.

Operator	Name	Description	Example	Try it
+	Addition	Adds together two values	x + y	Try it »
-	Subtraction	Subtracts one value from another	x - y	Try it »
*	Multiplication	Multiplies two values	x * y	Try it »
/	Division	Divides one value by another	x / y	Try it »
%	Modulus	Returns the division remainder	x % y	Try it »
++	Increment	Increases the value of a variable by 1	++x	Try it »
--	Decrement	Decreases the value of a variable by 1	--x	Try it »

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Java Assignment Operators

Assignment operators are used to assign values to variables.

In the example below, we use the **assignment** operator (=) to assign the value **10** to a variable called **x**:

Example

```
int x = 10;
```

[Try it Yourself »](#)

The **addition assignment** operator (+=) adds a value to a variable:

Example

```
int x = 10; x += 5;
```

[Try it Yourself »](#)

A list of all assignment operators:

Operator	Example	Same As	Try it
=	x = 5	x = 5	Try it »
+=	x += 3	x = x + 3	Try it »
-=	x -= 3	x = x - 3	Try it »
*=	x *= 3	x = x * 3	Try it »
/=	x /= 3	x = x / 3	Try it »
%=	x %= 3	x = x % 3	Try it »
&=	x &= 3	x = x & 3	Try it »
=	x = 3	x = x 3	Try it »
^=	x ^= 3	x = x ^ 3	Try it »
>>=	x >>= 3	x = x >> 3	Try it »
<<=	x <<= 3	x = x << 3	Try it »

x += 3

x = x + 3

==

x > y

x < y

>=

<=

Java Comparison Operators

Comparison operators are used to compare two values:

Operator	Name	Example	Try it
==	Equal to	x == y	Try it »
!=	Not equal	x != y	Try it »
>	Greater than	x > y	Try it »
<	Less than	x < y	Try it »
>=	Greater than or equal to	x >= y	Try it »
<=	Less than or equal to	x <= y	Try it »

Java Logical Operators

Logical operators are used to determine the logic between variables or values:

Operator	Name	Description	Example	Try it
&&	Logical and	Returns true if both statements are true	x < 5 && x < 10	Try it »
	Logical or	Returns true if one of the statements is true	x < 5 x < 4	Try it »
!	Logical not	Reverse the result, returns false if the result is true		

~(T) = F

a	b	a & b	a b
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    int x=8;  
    System.out.println(x++);  
    System.out.println(x);  
}
```

x = 8

++ x [post inc]
-- x [post dec]

x-- → 8

x → 7

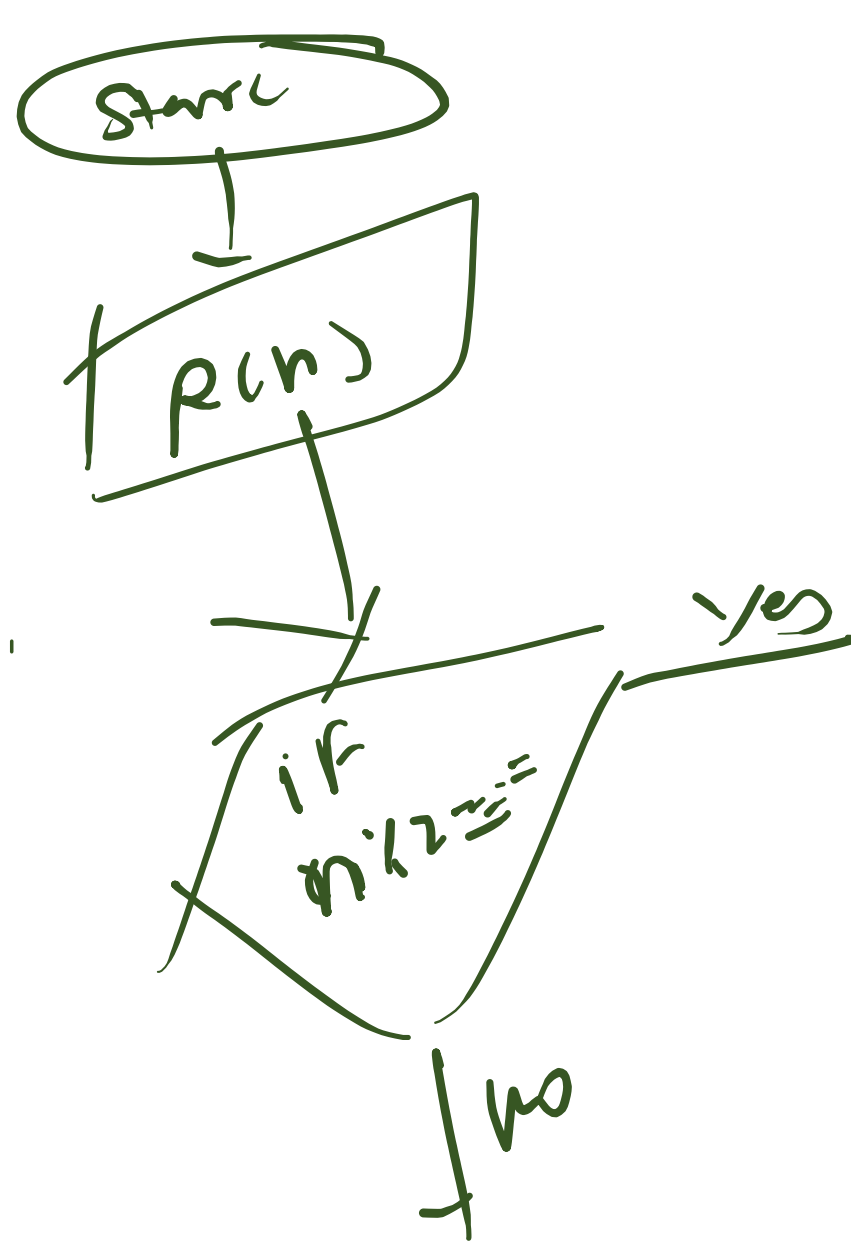
x = 8 9 8 9 8

```
int c = ++x - --x + x++ - x--;
```

9 - 8 + 8 - 9

x = 8 9 8 9 8

```
int c = ++x + x-- + x++ + --x;  
9 + 9 + 8 + 8  
18 + 16 = 34
```



① int m = 67

m =

m >= 75
→ use A
m >= 65 and m < 75 → B
m >= 55 and m < 65 → C
m >= 45 and m < 55 → pass
m < 45 → fail

⑩ n ≥ 100
⑪ n ≥ 282 83 n ≤ 473 macbook
⑫ n > 50 n ≤ 93 cycle
⑬ n ≥ 789 n ≤ 989 kolkata
⑭ But use next time
⑮ n ≤ 200 Bike
n ≤ 150 → hero
→ ktm
310 ≥ m1 and m2
80. ATLS
→ nuclea
800 Red green